

Power BI & DAX Quick Notes

Quick Revision Notes · by Mahendra Singh · CrackAnalytics

Architecture flow

Sources → Power Query (ETL) → Data Model (star schema) → DAX measures → Visuals → Publish to Service → Gateway + Scheduled refresh → Apps/Workspaces.

Star schema rules

- Central fact (transactions + numbers) surrounded by dimensions (who/what/when/where)
- Single-direction relationships (dim → fact), integer keys, mark the Date table
- Star vs Snowflake: denormalized (fast, PBI-preferred) vs normalized sub-dimensions

Storage modes

- **Import** — in-memory VertiPaq, fastest, full DAX (default choice)
- **DirectQuery** — live queries, huge/real-time data, slower visuals, limited DAX
- **Composite/Dual** — import dims + DQ facts, best of both
- **Direct Lake (Fabric)** — reads OneLake parquet at near-import speed, no refresh copies

DAX essentials

```
Total Sales    = SUM(Sales[Amount])
Profit %       = DIVIDE([Profit],[Total Sales])  -- safe divide
Sales LY       = CALCULATE([Total Sales],
                          SAMEPERIODLASTYEAR('Date'[Date]))
YoY %          = DIVIDE([Total Sales]-[Sales LY],[Sales LY])
Running Total  = CALCULATE([Total Sales],
                          FILTER(ALLSELECTED('Date'),'Date'[Date]<=MAX('Date'[Date])))
% of Total     = DIVIDE([Total Sales],
                          CALCULATE([Total Sales], ALL(Sales)))
```

- **Measure vs calculated column** — query-time & dynamic vs stored per-row (avoid when possible)
- **SUM vs SUMX** — column aggregate vs row-by-row iterator (Qty*Price)
- **ALL vs ALLSELECTED vs ALLEXCEPT** — remove all filters / respect slicers / keep listed columns
- **VALUES vs SELECTEDVALUE** — table of distincts vs single selection (else alternate)
- **Filter context** — the active filters on a cell; CALCULATE modifies it
- **TREATAS** — virtual relationship: filter a table with values from an unrelated one

Performance checklist

- Remove unused & high-cardinality columns · measures over calc columns
- Query folding in Power Query (View Native Query) · incremental refresh (RangeStart/RangeEnd)
- 5–10 visuals max per page · page filters before visual filters
- Diagnose with Performance Analyzer FIRST, then fix the top offender

RLS (Row-Level Security)

- Static role: [Region] = "West" — one role per value
- Dynamic: user table + [Email] = USERPRINCIPALNAME() — one rule for all users
- Manage Roles in Desktop → test View As Role → assign members in Service